Sprinkle the Salt: RO-Combiners with $\mathcal{O}(n)$ Random-Bits

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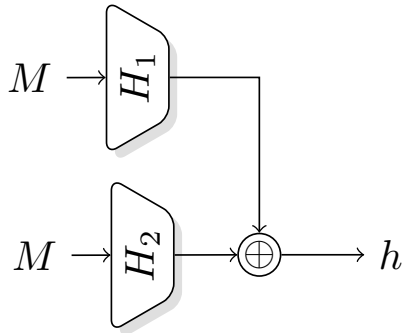
Main Message

- RO-Combiners is an hash indifferentiability problem.
  - Classical indifferentiability approaches are seemingly working.
- RO-Combiner security is a two stage game.
  - Lots of Definition Nuances.

Today's Focus

- RO-Combiners is an hash indifferentiability problem.
  - Classical indifferentiability approaches seemingly works.

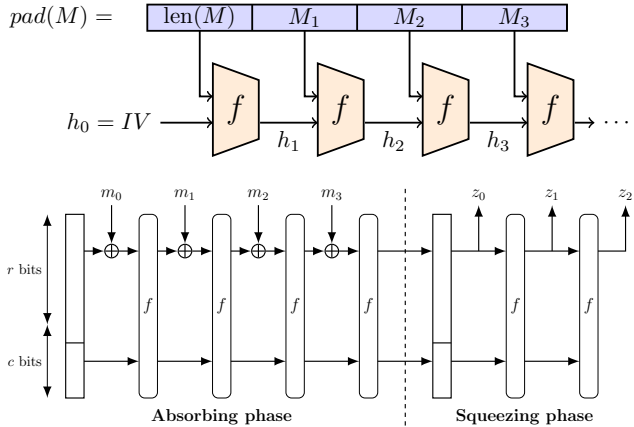
XOR is a RO-Combiner for independent hashes



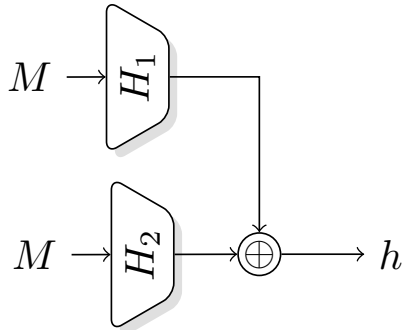
H_1, H_2 are independent.

H_b is Random Oracle $\implies$ $H_1 \oplus H_2$ is Random Oracle

Hash Functions are Built from Fixed-Input-Length Primitives



XOR is a RO-Combiner



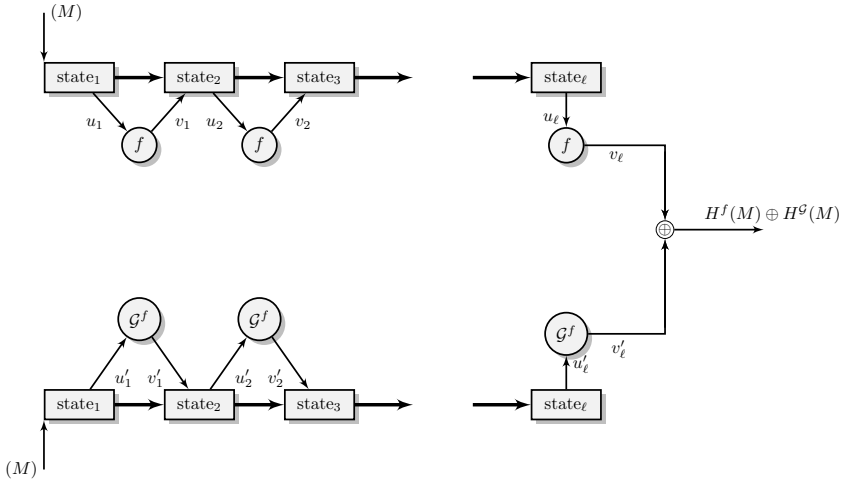
What happens if

- The hashes are built from smaller primitives, say f_1 and f_2?
- f_1, f_2 are NOT independent

How to get RO-Combiner?

What about the following?

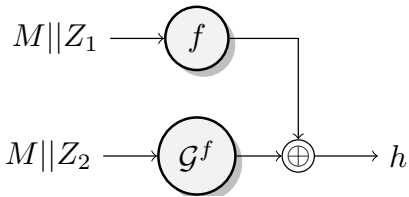
XOR of Generalised Domain Extensions



XOR of Fixed-Input-Length RO [DFGHP23]

Construction must be salted

Independent, uniform random salts Z_1, Z_2 for each hash evaluation.



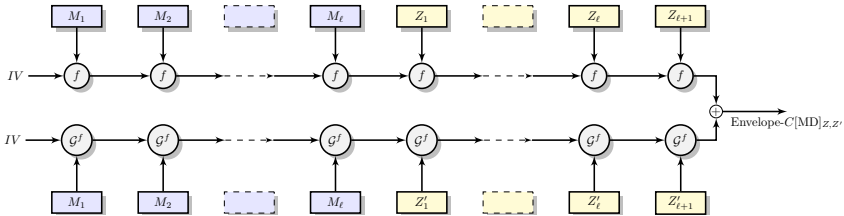
Parameters for n-bit security

$\|Z_i\| \geq \|M\| + n$

Insight

$\mathcal{G}(M, Z_2)$ can not query $f(\cdot, Z_1)$.

Fixed-Input-Length Combiner [DFGHP23, DGH25]



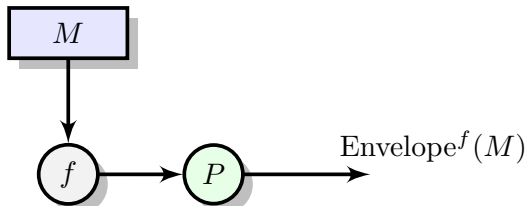
Parameters for n-bit security

$\|Z\| \geq \|M\| + n$

Our Approach

Envelope Construction [DRS2009]

Preimage-Aware Hash followed by *independent* Postprocessor



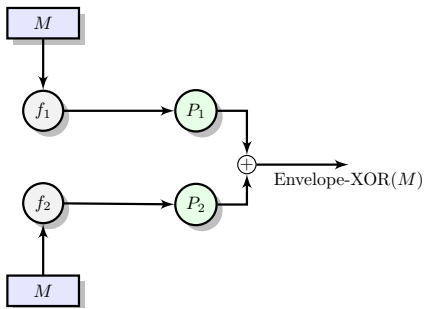
Simulating P

Simulator for P, on input x, retrieves a message M, $x = f(M)$, sets $P(x) = \mathcal{F}(M)$

Our Approach: Extending Standard Indifferentiability

XOR of two *independent* postprocessor EMD evaluation

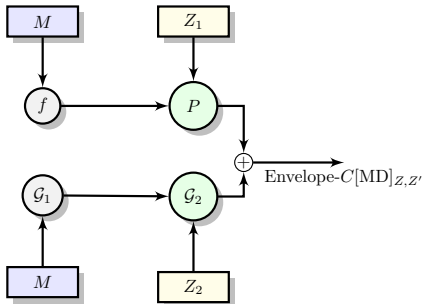
Simulator for P_1, on input x, retrieves a message M s.t. $x = f_1(M)$, sets $P(x) = \mathcal{F}(M) \oplus P_2(f_2(M))$



Our Approach: Extending Standard Indifferentiability

When the other hash is not independent

Use a Large post-processor! $\|Z_i\|$ is n-bit larger than $\mathcal{G}_1$ input.



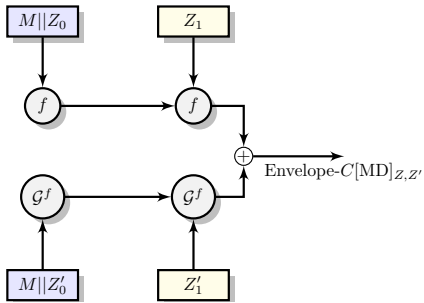
Insight

$\mathcal{G}_1(M), \mathcal{G}_2(Z_2)$ can not query $P(\cdot, Z_1)$.

Our Approach: Extending Standard Indifferentiability

When we have a single f and the other hash is not independent

Restrict the input-size of $\mathcal{G}$! $3n/4$-bit Z_0.



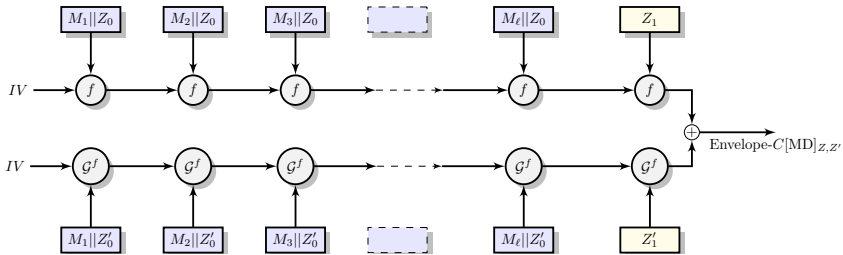
Insight

$\mathcal{G}_1(M, Z_0), \mathcal{G}_2(., Z_2)$ can not query $f(\cdot, Z_1)$.

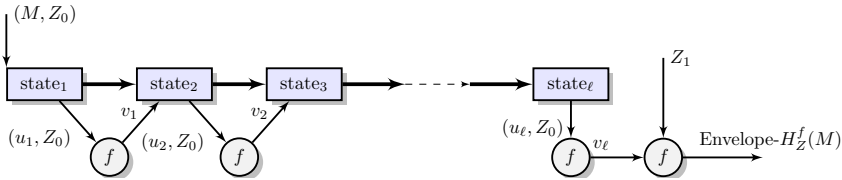
Combiner from standard MD

How to get “fresh” final input with a single f?

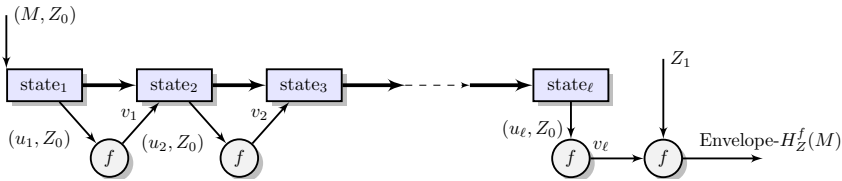
Brined-HAIFA (salt-soaked-HAIFA) style padding



Brined-HAIFA for Generic Design



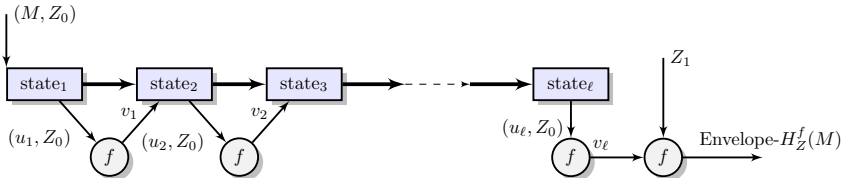
Brined-HAIFA for Generic Design



Future Work

Combiner for Sponge and other Permutation-Based Designs.

Brined-HAIFA for Generic Design



Future Work

Combiner for Sponge and other Permutation-Based Designs.

Thank You